

Algorithms and Data Structures

Problem Sheet 9

Hermann Schichl

100. Write pseudocode for deleting a node of a B-Tree.
101. The diameter of a tree $T = (V, E)$ is defined as $\max\{\delta(u, v) \mid u, v \in V\}$, that is, the largest of all shortest-path distances in the tree. Give an efficient algorithm to compute the diameter of a tree, and analyze the running time of your algorithm.
102. Show that in a directed graph, edge (u, v) is
- (a) a tree edge or forward edge if and only if $u.d < v.d < v.f < u.f$,
 - (b) a back edge if and only if $v.d \leq u.d < u.f \leq v.f$,
 - (c) a cross edge if and only if $v.d < v.f < u.d < u.f$.